

Week 8 Progress Summary

- - Completed a non-disruptive backend reliability contribution by implementing GET /health in newBackend/BackendCode/app.js.
- - Verified reliability behavior through smoke testing (newBackend/tests/api.smoke.test.js), including health endpoint response checks.
- - Updated architecture documents to reflect implemented changes, including /health status in docs/HLD.md and docs/LLD.md.
- - Built and organized evidence artifacts for traceability: worklog, evidence index, snippet showcase, planner mapping, and SFIA alignment.
- - Pushed Week 8 technical and documentation updates to local Git branch for versioned tracking and team visibility.
- - Completed software architecture certification activity and linked learnings to architecture-focused technical contribution work.

SFIA Alignment (Week 8)

- **Data analytics (DAAN):** correlation behavior analysis and data flow interpretation.
- **Information security (SCTY):** reliability and validation controls via health checks and safe request handling.
- **Functional testing (TEST):** smoke-test suite proving endpoint stability.
- **Systems integration and build (SINT):** architecture mapping across frontend, backend, and analytics paths.
- **Knowledge management (KNOW):** evidence index, planner traceability, and structured documentation.

Reliability Evidence Infographic

Reliability Engineering Evidence Snapshot

Implemented Controls

- GET /health endpoint
- app.js + server.js split for testability
- Repository path resolution hardening
- npm start / npm test scripts

Files:

newBackend/BackendCode/app.js
newBackend/BackendCode/repositories/mockRepository.js

Smoke Test Results

Test suite: newBackend/tests/api.smoke.test.js

PASS 1: GET /

PASS 2: GET /health

PASS 3: GET /api/stream-names

PASS 4: POST /api/filter-streams invalid payload

Overall: 4 / 4 passed

Why This Is Non-Disruptive

- Existing API contracts are preserved.
- Health endpoint is additive and does not change business logic routes.
- Test coverage adds validation evidence without runtime behavior changes.
- Path hardening improves startup reliability under WSL/dev contexts.

Proof links: docs/evidence/worklog.md, docs/evidence/evidence_index.md

Reliability Engineering Evidence

Week 8 Technical Snippets

Technical Contribution Evidence (Code Snippets)

1) Reliability Endpoint (Non-Disruptive Additive Control)

Source: newBackend/BackendCode/app.js

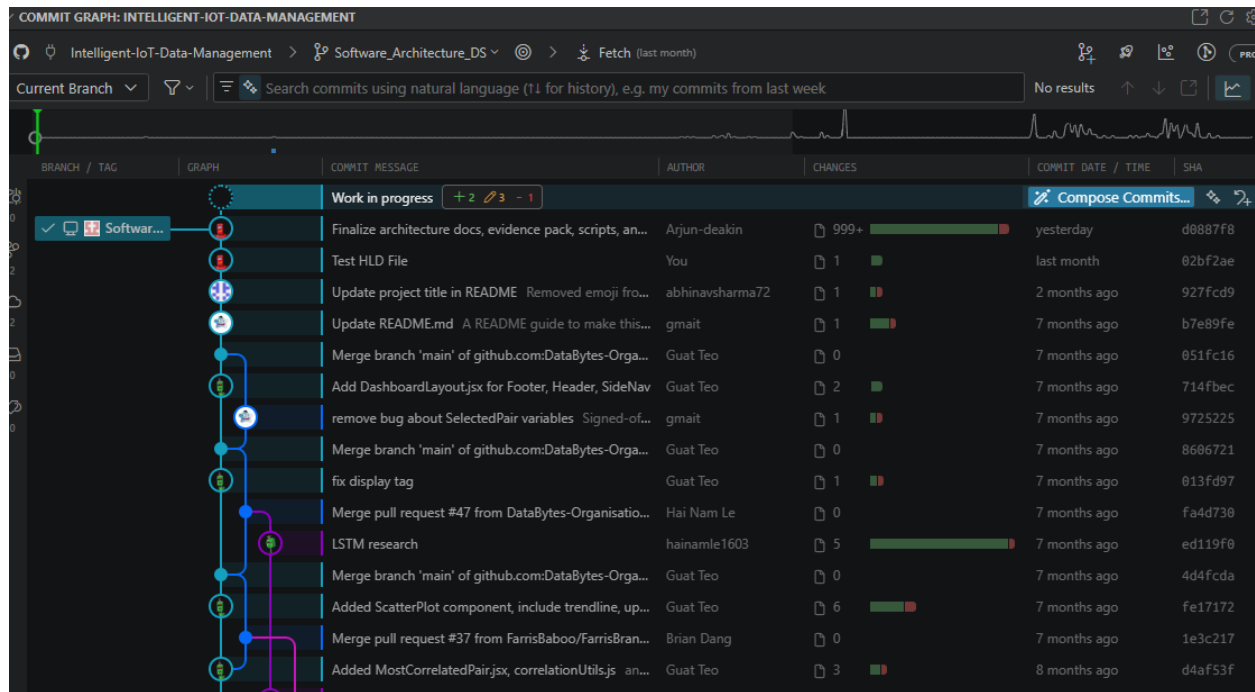
```
app.get('/health', (req, res) => {  
  res.status(200).json({  
    status: 'ok',  
    timestamp: new Date().toISOString(),  
    uptimeSeconds: process.uptime()  
  });  
});
```

```
JS apismoke.test.js JS app.js X
newBackend > BackendCode > JS app.js > app.get('/health') callback
  Arjun-deakin, last week | 1 author (Arjun-deakin)
1  const path = require('path');
2  const express = require('express');
3  const cors = require('cors');
4  const dotenv = require('dotenv');
5
6  dotenv.config({ path: path.resolve(__dirname, '../.env') });
7
8  const mockRoutes = require('./routes/mock');
9
10 const app = express();
11
12 app.use(cors());
13 app.use(express.json());
14
15 app.get('/', (req, res) => {
16   res.send('Backend is running');
17 });
18
19 app.get('/health', (req, res) => {
20   res.status(200).json({
21     status: 'ok',
22     timestamp: new Date().toISOString(),
23     uptimeSeconds: process.uptime()
24   });
25 });
26
27 app.use('/api', mockRoutes);
28
29 module.exports = app;
30
```

2) Automated Smoke Test Verification

```
JS apismoke.test.js X JS app.js
newBackend > tests > JS apismoke.test.js > test('GET / returns backend status text') callback
  Arjun-deakin, last week | 1 author (Arjun-deakin)
1  const test = require('node:test');
2  const assert = require('node:assert/strict');
3
4  const app = require('../BackendCode/app');
5
6  let server;
7  let baseUrl;
8
9  test.before(() => {
10   server = app.listen(0);
11   const address = server.address();
12   baseUrl = `http://127.0.0.1:${address.port}`;
13 });
14
15 test.after(() => {
16   server.close();
17 });
18
19 test('GET / returns backend status text', async () => {
20   const response = await fetch(`${baseUrl}/`);
21   const text = await response.text();
22
23   assert.equal(response.status, 200);
24   assert.equal(text, 'Backend is running');
25 });
26
27 test('GET /health returns service health payload', async () => {
28   const response = await fetch(`${baseUrl}/health`);
29   const body = await response.json();
30
31   assert.equal(response.status, 200);
32   assert.equal(body.status, 'ok');
33   assert.equal(typeof body.timestamp, 'string');
34   assert.equal(typeof body.uptimeSeconds, 'number');
35 });
36
37 test('GET /api/stream-names returns non-empty stream array', async () => {
38   const response = await fetch(`${baseUrl}/api/stream-names`);
39   const body = await response.json();
40
41   assert.equal(response.status, 200);
42   assert.ok(Array.isArray(body));
43   assert.ok(body.length > 0);
44 });
45
46 test('POST /api/filter-streams rejects invalid payload', async () => {
47   const response = await fetch(`${baseUrl}/api/filter-streams`, {
48     method: 'POST',
49     body: 'invalid payload'
50   });
51   const status = response.status;
52   const text = await response.text();
53   assert.equal(status, 400);
54   assert.equal(text, 'Invalid payload');
55 });
56
```

Git Evidences



Video Evidence

Recap: Call with RACHIT GOYAL Saturday, | Meeting | Microsoft Teams

Code Walkthrough

<https://deakin.au.panopto.com/Panopto/Pages/Viewer.aspx?id=11af56f7-8287-4475-8278-b43f006d6395>

Code Demo

<https://deakin.au.panopto.com/Panopto/Pages/Viewer.aspx?id=31d7fe82-457c-4fa1-8235-b445005cc618>

Planner Governance Evidence (Software Architecture Bucket)

Planner board used for architecture task planning and progress traceability:

- <https://planner.cloud.microsoft/webui/plan/Zx8AozAXI0qGNjreoURensgADRUx/view/board?tid=d02378ec-1688-46d5-8540-1c28b5f470f6>

Planner Snippets

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planner.cloud.microsoft/webui/mytasks/assignedtome/view/board?tid=d02378ec-1688-46d5-8540-1c28b5f470f6

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☰

DEAKIN UNIVERSITYPlanner

📅

My Day

📌

My Tasks

📅

My plans

📌

Fixed

📌

Fixed items will appear here.

🏠

My Tasks

📅

Grid

📌

Board

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AllPrivate tasksAssigned to meFlagged emails

Not started

System ArchitectureData ScienceLive

Intelligent IoT Data Management

ARCH-015: Prepare for Implementation Phase

Description

Supporting transition from planning/design activities towards implementation and integration phases of the IoT

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📅 15/05

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In progress

System ArchitectureLive

Intelligent IoT Data Management

ARCH-013: Setup Release Notes Process

Description

Preparing an initial release notes and documentation tracking process to improve visibility of architecture and implementation

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📅 15/05

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System ArchitectureActive

Intelligent IoT Data Management

ARCH-014: Conduct Architecture Review

Description

Conducting architecture review activities to validate system structure, integration flow, and implementation alignment across

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📅 15/05

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🔴 System Architecture

Intelligent IoT Data Management

ARCH-006: Document Integration Points

Description

Currently documenting integration points between frontend, backend, analytics, and data processing components to improve

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📅 15/05

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🔴 System Architecture

Intelligent IoT Data Management

ARCH-012: Define Architecture Governance

Description

Working on defining architecture governance practices to improve consistency in system structure

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📅 15/05

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Completed

System Architecture

Intelligent IoT Data Management

ARCH-004: Finalize High-Level Design Draft

Team: System Architecture

Resources: Handover doc, team inputs

Description: Finalize overall system architecture and align with teams

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📅 Completed by ARJUN SINGH on...

System ArchitectureData Science

Intelligent IoT Data Management

ARCH-003: Define End-to-End Data Flow

Team: System Architecture

Resources: Team inputs

Description: Design data flow from IoT sensors → ML → Backend → UI

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📅 Completed by ARJUN SINGH on...

System ArchitectureData Science

Intelligent IoT Data Management

ARCH-005: Conduct Team Alignment Session

Team: System Architecture

Resources: H&D document

Description: Align ML, Backend, UI teams on architecture

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📅 Completed by ARJUN SINGH on...

System Architecture

Intelligent IoT Data Management

ARCH-006: Start Low-Level Design (SDD)

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📅 Completed by ARJUN SINGH on...

System Architecture

Intelligent IoT Data Management

ARCH-010: Product Owner Alignment

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+ New plan

▼

Certificate copy of the Course



Project Worklog

PROJECT_HOURS_LOG_WEEK1_8.docx

Research References Used

1. <https://www.linkedin.com/learning/software-architecture-foundations/the-importance-of-software-architecture?u=2104084>
2. <https://lms.simplilearn.com/courses/8356/Introduction-to-Software-Architecture>.
3. Deakin AI and Literature Review Guide
<https://deakin.libguides.com/ai-literature-review>